Cohomological full faithfulness for the birhizomic affine fan

The result

We prove that the answer is yes. Throughout, sheaves on A_P and B_P mean sheaves on the sites A_P^{op} and B_P^{op}. Since the topology on B_P^{op} is trivial, a sheaf on B_P is the same thing as a presheaf on B_P^{op}, equivalently a covariant functor

$$B_P \longrightarrow \mathbf{Ab}.$$

Theorem 1. *Let P be an integral, saturated, sharp commutative monoid. Let*

$$\pi^{-1} : \mathrm{Sh}(B_P) \longrightarrow \mathrm{Sh}(A_P)$$

be the functor obtained from the inclusion $B_P^{\mathrm{op}} \subset A_P^{\mathrm{op}}$ by left Kan extension and sheafification, and let π_* be its right adjoint. Then, for every presheaf of abelian groups F on B_P^{op}, the adjunction morphism

$$F \longrightarrow \mathrm{R}\,\pi_*\pi^{-1}F$$

is a quasi-isomorphism.

The proof uses the standard affine comparison theorem for valuative, equivalently birhizomic, fans. This theorem is the affine form of Kato's cohomological contractibility theorem for valuative fans; see Kato–Nakayama–Usui [**?**, Section 3], Kato [**?**, Sections 1–3], and Ogus [**?**, Chapter V, Sections 1–2]. In the language of sites it is also an application of the locally-aspherical comparison theorem of SGA 4, Exposé IX; see [**?**, Exposé IX, Theorem 5.7].

1 Faces and the left Kan extension

A face of P is a submonoid $S \subset P$ such that, whenever $a + b \in S$, then $a, b \in S$. If $x \in P$, define the principal face generated by x by

$$S_x := \{p \in P \mid p \leq nx \text{ for some } n \geq 0\}.$$

Equivalently, $S_x = \{p \in P \mid p \prec x\}$.

For a face $S \subset P$, write P/S for the sharp monoid obtained from P by making every element of S invertible and then sharpifying. Thus $P \to P/S$ is initial among homomorphisms from P which send every element of S to 0.

Lemma 2. *For every $x \in P$, the monoid $P[-x]^\sharp$ is canonically isomorphic to P/S_x.*

Proof. Let N be a sharp monoid and let $\varphi : P \to N$ be a homomorphism with $\varphi(x) = 0$. If $p \in S_x$, then $p + r = nx$ for some $r \in P$ and some $n \geq 0$. Hence

$$\varphi(p) + \varphi(r) = n\varphi(x) = 0.$$

Since N is sharp, $\varphi(p) = 0$. Thus every map killing x kills all of S_x. Conversely $x \in S_x$, so every map killing S_x kills x. The two universal properties are therefore the same. □

Hence, up to equivalence, the category B_P may be identified with the full subcategory of A_P consisting of the quotients P/S for S a principal face. There is a unique morphism

$$P/S \longrightarrow P/T$$

over P if and only if $S \subset T$.

For $Q \in A_P$, let

$$K_Q := \ker(P \to Q) = \{p \in P \mid p \mapsto 0 \text{ in } Q\}.$$

This is a face of P.

Let $F : B_P \to \mathbf{Ab}$ be a presheaf on B_P^{op}. Define a functor

$$\overline{F} : \mathrm{Face}(P) \longrightarrow \mathbf{Ab}$$

by

$$\overline{F}(S) := \mathrm{colim}_{S_x \subset S} F(P/S_x),$$

where the colimit is over the filtered category of principal faces contained in S. If $S = S_x$ is principal, then S_x is final in that indexing category, so

$$\overline{F}(S_x) \cong F(P/S_x).$$

Lemma 3. *Let $i : B_P \hookrightarrow A_P$ denote the inclusion. The presheaf left Kan extension of F along i is the presheaf*

$$Q \longmapsto \overline{F}(K_Q).$$

Consequently

$$\pi^{-1}F \cong a\big(Q \mapsto \overline{F}(K_Q)\big),$$

where a denotes sheafification for the birhizomic topology.

Proof. By the formula for a left Kan extension,

$$(\mathrm{Lan}_i F)(Q) = \mathrm{colim}_{(P/S_x \to Q)} F(P/S_x).$$

A morphism $P/S_x \to Q$ over P exists if and only if $S_x \subset K_Q$, and in that case it is unique. Therefore the indexing category is precisely the category of principal faces contained in K_Q, giving

$$(\mathrm{Lan}_i F)(Q) = \mathrm{colim}_{S_x \subset K_Q} F(P/S_x) = \overline{F}(K_Q).$$

Sheafifying gives the stated formula for $\pi^{-1}F$. □

2 The affine birhizomic comparison theorem

We use the following standard affine comparison theorem.

Theorem 4 (Affine valuative/birhizomic comparison)**.** *Let M be an integral, saturated, sharp monoid, and let*

$$E : \mathrm{Face}(M) \longrightarrow \mathbf{Ab}$$

be a covariant functor. Let E^{br} be the sheaf on A_M^{op} associated to the presheaf

$$R \longmapsto E\big(\ker(M \to R)\big).$$

Then the canonical map

$$E(\{0\}) \longrightarrow \mathrm{R}\,\Gamma(A_M^{\mathrm{op}}, E^{\mathrm{br}})$$

is an isomorphism in $D(\mathbf{Ab})$.

More generally, if $S \subset M$ *is a face, then applying the same statement to* M/S *gives*

$$E(S) \simeq \mathrm{R}\,\Gamma(A_{M/S}^{\mathrm{op}}, E_S^{\mathrm{br}}),$$

where E_S *is the induced functor on faces of* M/S*, i.e. on faces of* M *containing* S.

Reference and translation. This is the affine form of the cohomological contractibility theorem for the valuative fan of an affine fan. In Kato's language, $\mathrm{Spec}(M)_{\mathrm{val}} \to \mathrm{Spec}(M)$ is locally aspherical over every stratum. The birhizomic topology defined in the question is precisely the site-theoretic formulation of the valuative topology on affine fans: birhizomic morphisms are the finite presentation morphisms which satisfy the valuative lifting criterion, and birhizomic covers are the covers by valuative subdivisions. See Kato [**?**, Sections 1–3], Kato–Nakayama–Usui [**?**, Section 3], and Ogus [**?**, Chapter V, Sections 1–2]. The passage from local asphericity to the displayed derived global-section statement is exactly the comparison theorem for locally aspherical morphisms of sites, SGA 4, Exposé IX, Theorem 5.7 [**?**]. □

3 Proof of the theorem

Let $F : B_P \to \mathbf{Ab}$ be a presheaf on B_P^{op}. We must show that

$$F \longrightarrow \mathrm{R}\,\pi_*\pi^{-1}F$$

is a quasi-isomorphism.

Because the topology on B_P^{op} is trivial, it is enough to check this objectwise. Thus fix $x \in P$, and put

$$S := S_x, \qquad b_x := P/S = P[-x]^\sharp.$$

The right adjoint π_* is restriction to B_P. Therefore, for a sheaf G on A_P^{op},

$$(\mathrm{R}\,\pi_*G)(b_x) \simeq \mathrm{R}\,\Gamma\big((A_P^{\mathrm{op}})/b_x, G\big).$$

This is the standard formula identifying the derived functors of evaluation at an object with cohomology of the slice site; see SGA 4, Exposé V, Proposition 5.1 [**?**].

Now

$$(A_P^{\mathrm{op}})/b_x \simeq A_{b_x}^{\mathrm{op}} = A_{P/S}^{\mathrm{op}},$$

because an object of the slice is a morphism $b_x \to Q$ in A_P, equivalently a monoid homomorphism $P/S \to Q$.

By the computation of the left Kan extension above,

$$\pi^{-1}F \cong a\big(Q \mapsto \overline{F}(K_Q)\big).$$

Restricting to the slice $A_{P/S}^{\mathrm{op}}$, this becomes the sheaf associated to the presheaf

$$Q \longmapsto \overline{F}(K_Q),$$

where now K_Q is viewed as a face of P containing S. Equivalently, it is the sheaf E^{br} associated to the functor

$$E : \mathrm{Face}(P/S) \longrightarrow \mathbf{Ab}, \qquad T/S \longmapsto \overline{F}(T).$$

Applying Theorem **??** to the monoid P/S, we obtain

$$\mathrm{R}\,\Gamma\big(A^{\mathrm{op}}_{P/S}, (\pi^{-1}F)|_{A_{P/S}}\big) \simeq E(\{0\}) = \overline{F}(S).$$

Since $S = S_x$ is principal, S is final among the principal faces contained in S, and hence

$$\overline{F}(S) \cong F(P/S) = F(b_x).$$

Combining these identifications gives

$$(\mathrm{R}\,\pi_*\pi^{-1}F)(b_x) \simeq F(b_x),$$

concentrated in degree 0.

The construction is functorial in b_x, and the degree 0 map is exactly the adjunction unit

$$F(b_x) \longrightarrow (\pi_*\pi^{-1}F)(b_x).$$

Thus the adjunction morphism

$$F \longrightarrow \mathrm{R}\,\pi_*\pi^{-1}F$$

is an objectwise quasi-isomorphism. Since the topology on B^{op}_P is trivial, objectwise quasi-isomorphisms are precisely quasi-isomorphisms of complexes of sheaves on B^{op}_P. This proves the theorem.

References

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